Condition, Generating Functions, and Normal Distributions

Conditional Probability, Expectation and Variance / Generating Functions / Normal Distribution

The central question throughout this guide: *Given partial information, how can we compute probabilities, predict means and quantify risk? When aggregating independent random variables, how can we avoid repeated integration?*

1. **Conditioning:** Fix a variable to reduce the dimension of a complicated problem.
2. **Aggregation:** Reconstruct global answers from local results using total expectation and total variance.
3. **Generating functions:** Use derivatives of PGFs, MGFs and CGFs to derive means, variances and sums of independent variables.
4. **Gaussian geometry:** Obtain normal moments from MGFs, then explain conditional prediction and correlation through orthogonal projection.

TECHNICAL INTERVIEW REFERENCE

Foundations · Generating Functions · Gaussian Methods · Quant Interview Practice

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Learning path: master general methods before specific distributions, instead of stacking formulas by problem type.



Two unifying threads:

1. From *local conditional laws* to *overall means and variances*: Tower Property and EVVE.
2. From *transforms and derivatives* to *means, variances and full distributions*: PGF / MGF / CGF, with the normal distribution as a representative application.

Bridge problem: Derive the random-trade-count profit $S = \sum_{i=1}^{N} R_i$ by both methods, then show that the answers agree algebraically.

1 The Big Picture: Conditioning, Generating Functions, and Gaussian Models

1.1 From Conditioning to Aggregation and Transforms

Multivariable probability questions in Quant Researcher / Trader interviews may look unrelated, but many require only three steps: **condition first, average second, and choose the most economical representation last**.

$$\underbrace{f_{X,Y}(x,y)}_{\text{joint model}} \xrightarrow[\text{divide by marginal density}]{\text{condition}} \underbrace{f_{X|Y}(x \mid y)}_{\text{local law}} \xrightarrow[\text{weight and integrate}]{\text{aggregate}} \underbrace{f_X(x),\ \mathbb{E}[X],\ \operatorname{Var}(X)}_{\text{global law}}$$

For sums of independent variables, counts, or high-order normal moments, a second representation is useful: **a generating function maps a distribution to a differentiable function, and an independent sum becomes a product**. We therefore establish conditioning and aggregation first, then PGF / MGF / CGF in Part II, and finally apply them to Gaussian distributions, chi-square variables, signal extraction and interview questions. **Generating functions are a prerequisite for later derivations, not an afterthought.**

1.2 Discrete and Continuous: One Language of Probability Mass

The density $f_Y(y)$ is not the probability of the event $Y = y$; the latter is zero for a continuous distribution. The infinitesimal probability mass is

$$\Pr(y \le Y < y + \mathrm{d}y) \approx f_Y(y)\,\mathrm{d}y.$$

Thus, once conditioning on Y yields a function $h(y)$, the unconditional average is always $\mathbb{E}[h(Y)]$:

$$\boxed{\mathbb{E}[h(Y)] = \begin{cases} \sum_y h(y)\Pr(Y=y), & Y \text{ discrete}, \\ \int h(y) f_Y(y)\,\mathrm{d}y, & Y \text{ continuous}. \end{cases}}$$

Concept	Discrete form	Continuous form
Local probability mass	$\Pr(Y=y)$	$f_Y(y)\,\mathrm{d}y$
Aggregation operator	$\sum_y$	$\int \cdot\,\mathrm{d}y$
Conditional normalization	Divide by $\Pr(Y=y)$	Divide by $f_Y(y)$

> **Intuition and Memory**
>
> **The only translation dictionary you need:** The discrete sum $\sum$ and continuous integral $\int$ both *add contributions from conditional branches*; each contribution needs a weight. For $Y \sim \text{Uniform}(0,1)$, the weight happens to be $f_Y(y) = 1$: it is invisible in the formula, but never absent.

1.3 A Unified Integral Notation for Discrete and Continuous Cases

Write the distribution of Y as $P_Y(\,\mathrm{d}y)$. Then all mixed cases have the same form:

$$\Pr(A) = \int \Pr(A \mid Y=y) P_Y(\,\mathrm{d}y), \quad \mathbb{E}[X] = \int \mathbb{E}[X \mid Y=y] P_Y(\,\mathrm{d}y).$$

These are **one pair of formulas, not two different theories to memorize**. In calculations, expand $P_Y(\,\mathrm{d}y)$ into either $\Pr(Y=y)$ or $f_Y(y)\,\mathrm{d}y$.

2 Conditional Distributions: When to Divide and When to Multiply

2.1 Six Fundamental Formulas Form a Closed Loop

Let X be the target variable and Y the conditioning variable. What matters most in the following table is not notation but the **direction of each operation**.

Operation	Discrete	Continuous
Marginal	$p_Y(y) = \sum_x p_{X,Y}(x,y)$	$f_Y(y) = \int f_{X,Y}(x,y)\,\mathrm{d}x$
Conditional distribution	$p_{X\mid Y}(x \mid y) = \dfrac{p_{X,Y}(x,y)}{p_Y(y)}$	$f_{X\mid Y}(x \mid y) = \dfrac{f_{X,Y}(x,y)}{f_Y(y)}$
Product rule	$p_{X,Y} = p_{X\mid Y} p_Y$	$f_{X,Y} = f_{X\mid Y} f_Y$

Operation	Discrete	Continuous
Total probability / marginalize	$p_X(x) = \sum_y p_{X\mid Y}(x \mid y)p_Y(y)$	$f_X(x) = \int f_{X\mid Y}(x \mid y)f_Y(y)\,\mathrm{d}y$
Conditional expectation	$\mathbb{E}[X \mid Y = y] = \sum_x x\, p_{X\mid Y}(x \mid y)$	$\mathbb{E}[X \mid Y = y] = \int x f_{X\mid Y}(x \mid y)\,\mathrm{d}x$
Total expectation	$\mathbb{E}[X] = \sum_y \mathbb{E}[X \mid Y = y]p_Y(y)$	$\mathbb{E}[X] = \int \mathbb{E}[X \mid Y = y]f_Y(y)\,\mathrm{d}y$

2.2 Conditioning as Slice Normalization

Fix $y = y_0$ on the joint density surface $f_{X,Y}(x, y)$. The resulting slice along x is $h(x) = f_{X,Y}(x, y_0)$, whose area is $\int h(x)\,\mathrm{d}x = f_Y(y_0)$, *not necessarily* 1. The valid conditional density is therefore

$$\boxed{f_{X\mid Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}}, \quad f_Y(y) = \int f_{X,Y}(u, y)\,\mathrm{d}u > 0.$$

This is a **normalization** operation, not merely a mechanical use of Bayes' formula. To compute $f_{Y\mid X}(y \mid x)$ instead, the denominator becomes $f_X(x)$. **The subscript identifies whose density it is; the arguments specify the evaluation point.**

> **Common Pitfalls**
>
> For continuous Y, strictly speaking $\mathbb{E}[X \mid Y = y]$ denotes a version of conditional expectation defined almost everywhere with respect to the law of Y. It is not the elementary fraction $\Pr(A \cap \{Y = y\})/\Pr(Y = y)$ for a zero-probability event. When joint densities exist and $f_Y(y) > 0$, slice normalization gives the usual computational version.

2.3 Continuous Bayes' Theorem: From Normalized Slices to Updating

Above we derived a conditional density from the joint density $f_{X,Y}$. If instead we are given **a prior for latent X and an observation mechanism for Y conditional on X**, we use continuous Bayes' theorem. Assume the relevant densities exist and the observation y satisfies $0 < f_Y(y) < \infty$. Then

$$f_{X\mid Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{f_X(x)\, f_{Y\mid X}(y \mid x)}{f_Y(y)},$$
$$f_Y(y) = \int_{-\infty}^{\infty} f_X(u) f_{Y\mid X}(y \mid u)\,\mathrm{d}u.$$

Combining the two equations gives the complete continuous Bayes' formula:

$$\boxed{f_{X\mid Y}(x \mid y) = \frac{f_X(x)\, f_{Y\mid X}(y \mid x)}{\int_{-\infty}^{\infty} f_X(u) f_{Y\mid X}(y \mid u)\,\mathrm{d}u}.}$$

There are only three steps: (1) use the product rule to write the joint density as "prior $\times$ observation mechanism"; (2) integrate over the latent variable u to obtain the marginal *density* at the observation y; (3) divide by that density to normalize the posterior over x to one. Bayes' theorem is thus **the same slice-normalization operation** introduced above, not an unrelated theory.

Bayes term	Notation	Statistical meaning
Prior	$f_X(x)$	Density of unknown X before observing data; integration variable x.
Likelihood	$f_{Y\|X}(y \mid x)$	Density of observing y if $X = x$; with y fixed, a function of x.
Evidence	$f_Y(y)$	Integral of "prior $\times$ likelihood" over candidate values of x; the marginal density of Y.
Posterior	$f_{X\|Y}(x \mid y)$	Updated density for X after observing y; integrates to one over x.

Intuition and Memory

Remember: "updated belief $\propto$ prior belief $\times$ new information":

$$\underbrace{f_{X|Y}(x \mid y)}_{\text{Posterior}} \propto_x \underbrace{f_X(x)}_{\text{Prior}} \underbrace{f_{Y|X}(y \mid x)}_{\text{Likelihood}}.$$

The denominator does exactly one job: **normalize this kernel over unknown** x. The observation y is fixed, and the evidence does not depend on x. The Kernel Trick is precisely the practice of omitting this normalization constant while recognizing a posterior kernel.

Common Pitfalls

Three easily confused points: (1) $f_Y(y)$ is a *density*, not the probability $\Pr(Y = y)$ of a continuous point event; (2) the likelihood $f_{Y|X}(y \mid x)$ is *not necessarily a normalized density in* x and cannot itself be treated as the posterior; (3) integrate the denominator over the latent variable u, not over the observed value y.

The mixed case follows the same logic: If X is a discrete state and Y is a continuous observation, then

$$\Pr(X = k \mid Y = y) = \frac{\Pr(X = k)\, f_{Y|X}(y \mid k)}{\sum_j \Pr(X = j)\, f_{Y|X}(y \mid j)}.$$

When the unknown variable is discrete, the evidence is a sum, not an integral.

2.4 Gaussian Prior + Gaussian Likelihood: Derivation and Quant Example

Model: Suppose the true but unobserved asset alpha is

$$X \sim \mathcal{N}(\mu_0, \tau^2), \qquad Y = X + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, s^2), \quad \varepsilon \perp\!\!\!\perp X.$$

When $X = x$ is known, $Y \mid X = x \sim \mathcal{N}(x, s^2)$. Therefore,

$$\underbrace{f_X(x)}_{\text{Prior}} \propto_x \exp\left[-\frac{(x-\mu_0)^2}{2\tau^2}\right], \qquad \underbrace{f_{Y|X}(y \mid x)}_{\text{Likelihood}} \propto_x \exp\left[-\frac{(y-x)^2}{2s^2}\right].$$

Step 1: Continuous Bayes + Kernel Trick. Fix observed y. Since the evidence is constant with respect to x,

$$f_{X|Y}(x \mid y) \propto_x \exp\left[-\frac{1}{2}\left(\frac{(x-\mu_0)^2}{\tau^2} + \frac{(y-x)^2}{s^2}\right)\right].$$

Step 2: Collect the quadratic terms in x and complete the square.

$$\frac{(x-\mu_0)^2}{\tau^2}+\frac{(y-x)^2}{s^2}=\left(\frac{1}{\tau^2}+\frac{1}{s^2}\right)x^2 - 2\left(\frac{\mu_0}{\tau^2}+\frac{y}{s^2}\right)x+C(y).$$

The posterior is therefore also Gaussian, with parameters

$$X\mid(Y=y)\sim\mathcal{N}(\mu_{\text{post}},v_{\text{post}}),$$
$$\boxed{v_{\text{post}}}=\boxed{\left(\frac{1}{\tau^2}+\frac{1}{s^2}\right)^{-1}=\frac{\tau^2s^2}{\tau^2+s^2}},$$
$$\boxed{\mu_{\text{post}}}=\boxed{v_{\text{post}}\left(\frac{\mu_0}{\tau^2}+\frac{y}{s^2}\right)=\mu_0+\frac{\tau^2}{\tau^2+s^2}(y-\mu_0)}.$$

Moreover, the evidence in the total-probability calculation is the density $f_Y(y)$ of $Y\sim\mathcal{N}(\mu_0,\tau^2+s^2)$, not the probability of a point event.

Step 3: Economic intuition for the parameters. Define

$$K=\frac{\tau^2}{\tau^2+s^2}\in(0,1),\qquad \mu_{\text{post}}=(1-K)\mu_0+Ky,\quad v_{\text{post}}=(1-K)\tau^2.$$

This is the **Gain** $\times$ **Innovation** form: updated alpha estimate = prior estimate + reliability weight $K\times$ new information $(y-\mu_0)$. Equivalently, the posterior *precision* $1/v_{\text{post}}$ equals prior precision $1/\tau^2$ plus observation precision $1/s^2$.

As observation noise $s^2\to 0$, $K\to 1$, so the new signal is almost completely trusted; as $s^2\to\infty$, $K\to 0$, so the update reverts to the prior. In this Gaussian model, **posterior variance does not depend on the observed value** y, but is strictly smaller than prior variance.

Complete interview example (nonzero prior mean): Suppose $X\sim\mathcal{N}(1,4)$, independent noise $\varepsilon\sim\mathcal{N}(0,9)$, and the observation is $Y=X+\varepsilon=7$. Find $\mathbb{E}[X\mid Y=7]$ and $\text{Var}(X\mid Y=7)$.

$$K=\frac{4}{4+9}=\frac{4}{13},\qquad \boxed{\mathbb{E}[X\mid Y=7]=1+\frac{4}{13}(7-1)=\frac{37}{13}},$$
$$\boxed{\text{Var}(X\mid Y=7)=\frac{4\cdot 9}{4+9}=\frac{36}{13}}.$$

Because the noise is more dispersed than the alpha prior, the posterior moves only partway from 1 toward 7, rather than interpreting $Y=7$ as true alpha.

Step 4: Connect exactly to joint-normal OLS / projection. Independent normal variables remain jointly normal under linear transformation, so (X,Y) is jointly normal, with

$$\mathbb{E}[Y]=\mu_0,\qquad \text{Cov}(X,Y)=\tau^2,\qquad \text{Var}(Y)=\tau^2+s^2.$$

Gaussian conditioning therefore also gives

$$\mathbb{E}[X\mid Y=y]=\mu_0+\frac{\text{Cov}(X,Y)}{\text{Var}(Y)}(y-\mu_0),$$
$$\text{Var}(X\mid Y)=\tau^2-\frac{\text{Cov}(X,Y)^2}{\text{Var}(Y)}.$$

Substitution immediately gives the same μ_{post} and v_{post}. One route starts with **density updating**; the other starts with **covariance projection**; both yield the same posterior distribution. Multivariate Gaussian conditioning is the matrix version (see Section 9.2).

Intuition and Memory

One formula, two memory devices: In the Bayes view, Posterior $\propto$ Prior $\times$ Likelihood, with *precisions adding*. In the regression view, Posterior Mean = Prior Mean + Gain $\times$ Innovation. The two gains must agree:

$$\underbrace{\frac{\tau^2}{\tau^2+s^2}}_{\text{Bayesian weighting}} = \underbrace{\frac{\operatorname{Cov}(X,Y)}{\operatorname{Var}(Y)}}_{\text{Gaussian regression}}.$$

2.5 When to Divide and When to Multiply

1. **Joint distribution $\to$ local law: divide.** Given $f_{X,Y}$, if the question asks for the density, mean or probability of $X \mid Y = y$, find $f_Y(y)$ and divide $f_{X,Y}/f_Y$.
2. **Local law $\to$ global law: multiply and aggregate.** Given $\Pr(A \mid Y = y)$ or $f_{X|Y}$, if the target is $\Pr(A)$, f_X or $\mathbb{E}[X]$, multiply by $f_Y(y)\,\mathrm{d}y$ and integrate over y. For discrete Y, use $p_Y(y)$ and a sum instead.

Intuition and Memory

Kernel Trick (recognizing a probability kernel): While computing $X \mid Y = y$, the value y is fixed. Once the *support in* x is identified,

$$f_{X|Y}(x \mid y) \;\propto_x\; f_{X,Y}(x,y).$$

Recognizing the kernel in x (Gaussian, exponential, uniform, etc.) may give the conditional mean immediately. **The proportionality is with respect to x; never forget that the support may depend on y.**

2.6 Why Is Conditional Expectation Itself a Random Variable?

First compute a number for each fixed $Y = y$:

$$g(y) = \mathbb{E}[X \mid Y = y] = \int x f_{X|Y}(x \mid y)\,\mathrm{d}x.$$

After integrating out x, the resulting function g depends only on y. Before Y is observed, evaluate that function at the random variable to obtain

$$\boxed{\mathbb{E}[X \mid Y] = g(Y)}, \qquad \boxed{\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]}.$$

The second identity is the law of total expectation, or Tower Property. Verify directly:

$$\int \underbrace{\left[\int x f_{X|Y}(x \mid y)\,\mathrm{d}x\right]}_{g(y)} f_Y(y)\,\mathrm{d}y = \iint x f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y = \mathbb{E}[X].$$

> **Intuition and Memory**
>
> **The leftover-variable check:** When computing $g(y)$, you integrate with respect to $\mathrm{d}x$, so the result cannot still contain the dummy variable x. When evaluating $\mathbb{E}[g(Y)]$, integrate with respect to $\mathrm{d}y$; a constant final answer should contain no remaining dummy y. This quickly catches reversed inner and outer integrals.

2.7 Continuous, Discrete and Mixed Conditioning: Watch the Outer Variable

Conditioning variable	Target	Correct aggregation
Continuous Y	Discrete event A	$\Pr(A) = \int \Pr(A \mid Y = y) f_Y(y)\,\mathrm{d}y$
Continuous Y	Continuous target X	$f_X(x) = \int f_{X\mid Y}(x \mid y) f_Y(y)\,\mathrm{d}y$
Discrete N	Continuous payoff S	$\mathbb{E}[S] = \sum_n \mathbb{E}[S \mid N = n] \Pr(N = n)$
Discrete N	Discrete target X	$\Pr(X = k) = \sum_n \Pr(X = k \mid N = n) \Pr(N = n)$

Principle: the outer operator depends on the conditioning variable, while the inner operator depends on the target variable or event. If trade count N is discrete but each return R_i is continuous, you integrate over returns after conditioning on $N = n$, and finally sum over the probability masses of n.

3 From Means to Risk: Total Variance and Projection

3.1 The Law of Total Variance: Two Sources of Risk

Variance is defined as $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$. Set $m(Y) = \mathbb{E}[X \mid Y]$ and decompose the error:

$$X - \mathbb{E}[X] = \underbrace{X - m(Y)}_{\text{within-condition randomness}} + \underbrace{m(Y) - \mathbb{E}[X]}_{\text{variation of conditional means}}.$$

The cross term vanishes because

$$\mathbb{E}\big[(X - m(Y))h(Y)\big] = \mathbb{E}\big[h(Y)\mathbb{E}[X - m(Y) \mid Y]\big] = 0$$

(take $h(Y) = m(Y) - \mathbb{E}[X]$). Squaring and taking expectations yields

$$\boxed{\mathrm{Var}(X) = \underbrace{\mathbb{E}[\mathrm{Var}(X \mid Y)]}_{\text{EV: average conditional risk}} + \underbrace{\mathrm{Var}(\mathbb{E}[X \mid Y])}_{\text{VE: variation across conditional means}}}.$$

> **Intuition and Memory**
>
> **EVVE (Expected Variance + Variance of Expectation):**
>
> The first term asks "how dispersed is each group internally?" The second asks "how different are the means of the groups?" Compared with total expectation, total variance **adds the between-group difference**. This decomposition is especially valuable for random volatility, groups of default probabilities, and random trade counts.

3.2 Why Is It Called Orthogonal Projection?

Consider the inner-product space of centered random variables with finite second moments. Define $\langle U, V\rangle = \mathbb{E}[UV]$. Then

$$\|U\|_2^2 = \mathbb{E}[U^2], \qquad \mathrm{Cov}(X,Y) = \langle X - \mathbb{E}[X], Y - \mathbb{E}[Y]\rangle.$$

$\mathbb{E}[X \mid Y]$ is the L^2 orthogonal projection of X onto the space of square-integrable functions of Y:

$$\boxed{\mathbb{E}[(X - \mathbb{E}[X \mid Y])h(Y)] = 0.}$$

It also minimizes $\mathbb{E}[(X - g(Y))^2]$ over all measurable predictors $g(Y)$. **Conditional expectation is simultaneously a probabilistic average and the minimum-mean-square-error predictor.**

3.3 Regression Versus Conditioning: General Laws and Joint Gaussianity

The best *linear* predictor has the form $\widehat{X} = a + bY$, where $b = \mathrm{Cov}(X,Y)/\mathrm{Var}(Y)$. For an arbitrary distribution, however, the actual $\mathbb{E}[X \mid Y]$ need not be linear. Under **joint Gaussianity**, the linear predictor equals the conditional expectation exactly, and the residual is independent of Y. This bridge leads directly to Gaussian conditional distributions below.

4 Conditioning Archetypes: Five Essential Scenarios

4.1 Archetype A: Joint Density → Conditional Exponential

Problem: The joint density is

$$f_{X,Y}(x,y) = 2e^{-x-y}\mathbf{1}_{\{0<y<x<\infty\}}.$$

Find $\mathbb{E}[X \mid Y = y]$ and $\mathbb{E}[X]$.

Standard method Fix $y > 0$, noting that $x > y$:

$$\begin{aligned} f_Y(y) &= \int_y^\infty 2e^{-x-y}\,\mathrm{d}x = 2e^{-2y}, \\ f_{X|Y}(x \mid y) &= \frac{2e^{-x-y}}{2e^{-2y}} = e^{-(x-y)}\mathbf{1}_{\{x>y\}}, \\ \mathbb{E}[X \mid Y = y] &= \int_y^\infty x e^{-(x-y)}\,\mathrm{d}x = \int_0^\infty (u+y)e^{-u}\,\mathrm{d}u = y + 1. \end{aligned}$$

Interview shortcut: recognize the kernel For fixed y, the conditional kernel in x is proportional to e^{-x}, with support $x > y$. Therefore recognize immediately that

$$X \mid Y = y \overset{d}{=} y + W, \quad W \sim \mathrm{Exp}(1).$$

Also $Y \sim \mathrm{Exp}(2)$, so

$$\boxed{\mathbb{E}[X \mid Y = y] = y + 1, \qquad \mathbb{E}[X] = \mathbb{E}[Y] + 1 = \frac{3}{2}.}$$

Transferable insight: For any exponential kernel or truncated conditional distribution, inspect the *rate and support* first, rather than rushing into a double integral.

4.2 Archetype B: The Race Between Independent Exponential Waiting Times

Problem: Independent $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$. Find $\Pr(X < Y)$.

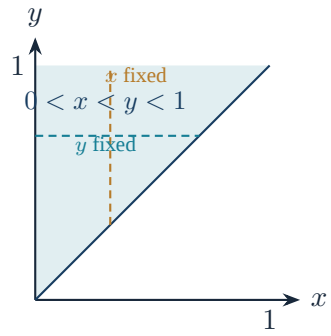
$$\begin{aligned}\Pr(X < Y) &= \int_0^\infty \Pr(X < y \mid Y = y)\, f_Y(y)\, \mathrm{d}y \\ &= \int_0^\infty (1 - e^{-\lambda_1 y})\lambda_2 e^{-\lambda_2 y}\, \mathrm{d}y \\ &= 1 - \frac{\lambda_2}{\lambda_1 + \lambda_2} = \boxed{\frac{\lambda_1}{\lambda_1 + \lambda_2}}.\end{aligned}$$

Intuition and Memory

Fix one variable and the other immediately becomes a CDF. For independent variables, $\Pr(X < Y \mid Y = y) = F_X(y)$. The same dimensional reduction handles $\Pr(X + Y < c)$ or $\Pr(XY < c)$: first rewrite the inequality as a one-dimensional condition for a fixed y.

4.3 Archetype C: Hierarchical Uniform Sampling—Support Before Integration

Problem: First draw $Y \sim \text{Uniform}(0, 1)$, then draw $X \mid Y = y \sim \text{Uniform}(0, y)$. Find $f_X(x)$, $\mathbb{E}[X]$ and $\text{Var}(X)$.



The conditional density is $f_{X|Y}(x \mid y) = \frac{1}{y}\mathbf{1}_{\{0<x<y\}}$. **When computing the marginal $f_X(x)$, x is fixed; valid conditions require $x < y < 1$:**

$$\boxed{f_X(x) = \int_x^1 \frac{1}{y}\, \mathrm{d}y = -\ln x, \quad 0 < x < 1.}$$

To find the mean, there is no need to integrate $-\ln x$ again:

$$\mathbb{E}[X \mid Y = y] = \frac{y}{2}, \qquad \boxed{\mathbb{E}[X] = \mathbb{E}\left[\frac{Y}{2}\right] = \frac{1}{4}}.$$

For variance, use EVVE with $\text{Var}(X \mid Y) = Y^2/12$, $\text{Var}(Y) = 1/12$, and $\mathbb{E}[Y^2] = 1/3$.

$$\boxed{\text{Var}(X) = \frac{1}{12}\frac{1}{3} + \frac{1}{4}\frac{1}{12} = \frac{1}{36} + \frac{1}{48} = \frac{7}{144}.}$$

Transferable insight: The support $0 < x < y < 1$ is not decoration: it changes the marginalization lower limit from 0 to x. Although the marginal density is $-\ln x$, conditioning is faster for its moments.

4.4 Archetype D: A Continuous Parameter Governs a Discrete Event

Problem: First draw the coin's heads probability $P \sim \text{Uniform}(0,1)$. Conditional on $P = p$, toss independently n times. Find the probability of all heads.

$$\Pr(\text{all heads} \mid P = p) = p^n, \qquad \boxed{\Pr(\text{all heads}) = \int_0^1 p^n \, \mathrm{d}p = \frac{1}{n+1}.}$$

Although the target is a discrete event, integration is over the *continuous conditioning variable* P. **The outer operation follows the condition; the inner operation follows the target.**

4.5 Archetype E: Profit from a Random Number of Trades (Preview)

Problem: Let $N \sim \text{Poisson}(\lambda)$. The returns R_i are i.i.d. with mean μ, variance σ^2, and independent of N. Set $S = \sum_{i=1}^{N} R_i$, with $S = 0$ when $N = 0$.

$$\mathbb{E}[S \mid N = n] = n\mu, \quad \text{Var}(S \mid N = n) = n\sigma^2.$$

By Tower and EVVE,

$$\boxed{\mathbb{E}[S] = \lambda\mu, \qquad \text{Var}(S) = \lambda\sigma^2 + \lambda\mu^2 = \lambda(\sigma^2 + \mu^2).}$$

The key distinction is between **intrinsic per-trade variation** $\lambda\sigma^2$ and **variation in the number of trades** $\lambda\mu^2$. The next part introduces generating functions, then derives the same result from $M_S(t) = G_N(M_R(t))$, showing why both conditional expectation and MGF methods solve the same problem.

5 Part II: Generating Functions as the Toolkit for Later Derivations

5.1 Why PGF, MGF and CGF?

A generating function hides the random variable in an exponent or a power. Differentiation then "pulls down" the desired factors X, X^2, or $X(X-1)$.

Name	Definition	When and why to use it
PGF	$G_X(s) = \mathbb{E}[s^X] = \sum_{k\geq 0} p_k s^k$	Nonnegative integer-valued X; probabilities and factorial moments; counts and random numbers of terms
MGF	$M_X(t) = \mathbb{E}[e^{tX}]$	Discrete or continuous when finite near zero; derivatives give raw moments; independent sums become products
CGF	$K_X(t) = \ln M_X(t)$	Cumulants: first derivative gives the mean, second the variance; independent sums become sums

For *nonnegative integer-valued* X, wherever the expressions are defined,

$$\boxed{M_X(t) = G_X(e^t)}, \qquad \boxed{G_X(s) = M_X(\ln s) \quad (s > 0)}.$$

Intuition and Memory

PGF and MGF are two faces of the same object: $s^X = e^{X \ln s}$. Substituting $s = e^t$ converts one into the other. The CGF then takes the *logarithm* of the MGF, making variance and independent sums easier to handle.

5.2 The Main Memory Device: Evaluate at Zero or One?

Our goal is to make the remaining exponential factor equal to 1 after differentiation.

$$G_X(1) = \mathbb{E}[1^X] = 1, \quad M_X(0) = \mathbb{E}[e^{0X}] = 1, \quad K_X(0) = \ln 1 = 0.$$

Remember: evaluate the base s at 1; for the base e^t, make the exponent zero by setting $t = 0$. This also explains why PGFs encode the probability *coefficients* of $X = 0, 1, 2, \ldots$, while MGFs near zero generate *moments*.

5.3 The Essential Derivative Table: Same First Moment, Different Second Moments

Function	Evaluation	First derivative	Second derivative and variance
PGF G	$s = 1$	$G'(1) = \mathbb{E}[X]$	$G''(1) = \mathbb{E}[X(X-1)]$; $\mathrm{Var}(X) = G''(1) + G'(1) - [G'(1)]^2$
MGF M	$t = 0$	$M'(0) = \mathbb{E}[X]$	$M''(0) = \mathbb{E}[X^2]$; $\mathrm{Var}(X) = M''(0) - [M'(0)]^2$
CGF K	$t = 0$	$K'(0) = \mathbb{E}[X]$	$K''(0) = \mathrm{Var}(X)$ (variance directly)

Why? Look at what differentiation does:

$$\frac{\mathrm{d}}{\mathrm{d}s} s^X = X s^{X-1}, \quad \frac{\mathrm{d}^2}{\mathrm{d}s^2} s^X = X(X-1) s^{X-2},$$
$$\frac{\mathrm{d}}{\mathrm{d}t} e^{tX} = X e^{tX}, \quad \frac{\mathrm{d}^2}{\mathrm{d}t^2} e^{tX} = X^2 e^{tX}.$$

Differentiating the *power* in the first line lowers its exponent by one each time; differentiating the *exponential* in the second line multiplies by another full factor of X each time.

The CGF uses the quotient rule:

$$K'(t) = \frac{M'(t)}{M(t)}, \qquad K''(t) = \frac{M''(t)M(t) - [M'(t)]^2}{[M(t)]^2}.$$

At $t = 0$, use $M(0) = 1$ to obtain $\mathrm{Var}(X)$ immediately.

Three stages of processing (a durable mnemonic)

CGF: the finished product. $K''(0)$ is the variance directly.
MGF: the semi-finished product. $M''(0)$ gives the second raw moment; subtract μ^2.
PGF: the unfinished foundation. $G''(1)$ is missing μ; add $G'(1)$, then subtract μ^2.

5.4 Why Higher PGF Derivatives Give Factorial Moments

For nonnegative integer-valued N,

$$G_N^{(r)}(1) = \mathbb{E}[N(N-1)\cdots(N-r+1)].$$

For Poisson(λ), the PMF contains $k!$; the falling factorial $k(k-1)\cdots$ cancels it down to $(k-r)!$. **This is fundamentally why PGFs work so well for count distributions.**

5.5 Why CGFs Are Especially Useful in Quant Finance

Independent $X_1, \ldots, X_n$ satisfy

$$M_{\sum_i X_i}(t) = \prod_i M_{X_i}(t), \qquad \boxed{K_{\sum_i X_i}(t) = \sum_i K_{X_i}(t)}.$$

Thus means and variances add naturally, as do higher-order *cumulants*. The CGF of a normal distribution is a quadratic polynomial,

$$K_{\mathcal{N}(\mu,\sigma^2)}(t) = \mu t + \frac{1}{2}\sigma^2 t^2,$$

so all third and higher cumulants vanish. **Caution:** Knowing only the mean and variance does not establish normality; the Gaussian signature is the absence of all higher-order terms in the CGF.

5.6 Derive Common Count and Waiting Distributions Before Gaussian Models

5.6.1 Poisson(λ): the Taylor Expansion of e^x

$$\begin{aligned} M_N(t) &= \sum_{k=0}^{\infty} e^{tk}\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} \\ &= \boxed{e^{\lambda(e^t-1)}}; \quad G_N(s) = \boxed{e^{\lambda(s-1)}}; \quad K_N(t) = \lambda(e^t-1). \end{aligned}$$

$K_N'(0) = \lambda$ and $K_N''(0) = \lambda$, hence $\mathbb{E}[N] = \mathrm{Var}(N) = \lambda$.

> **Intuition and Memory**
>
> **Three building blocks:** λ^k is the intensity factor for "wanting k events"; $e^{-\lambda}$ is the silent/no-extra-arrivals normalization baseline; and $1/k!$ removes duplicate arrival orderings. Multiplying by e^{tk} amounts to replacing λ with λe^t, so summing exponentials naturally produces $e^{\lambda(e^t-1)}$.

5.6.2 Exponential(λ): One Direct Integral

Here λ is the *rate*, and $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.

$$\begin{aligned} M_X(t) &= \int_0^{\infty} e^{tx}\lambda e^{-\lambda x}\,\mathrm{d}x = \frac{\lambda}{\lambda - t}, \quad t < \lambda, \\ K_X(t) &= \ln\lambda - \ln(\lambda - t). \end{aligned}$$

Therefore $\boxed{\mathbb{E}[X] = 1/\lambda,\ \mathrm{Var}(X) = 1/\lambda^2}$. Since the exponential distribution is continuous, its moments are *not* obtained from count-probability coefficients of a PGF $\mathbb{E}[s^X]$.

5.6.3 Binomial(n, p): Raise One Bernoulli Transform to the nth Power

A single coin toss $B \sim \text{Bernoulli}(p)$ has

$$M_B(t) = (1-p) + pe^t, \qquad G_B(s) = (1-p) + ps.$$

Since $N = \sum_{i=1}^{n} B_i$ is a sum of independent terms,

$$\boxed{M_N(t) = (1-p+pe^t)^n, \quad G_N(s) = (1-p+ps)^n, \quad K_N(t) = n\ln(1-p+pe^t)}.$$

Differentiation gives $\boxed{\mathbb{E}[N] = np, \quad \text{Var}(N) = np(1-p)}$. **No need to re-sum binomial coefficients $\binom{n}{k}$:** start with one toss and raise its transform to the nth power.

> **Intuition and Memory**
>
> **Why not PGF for a normal variable?** A PGF stores probabilities $\Pr(N = k)$ of a nonnegative integer-valued variable as *coefficients* of s^k. Continuous Gaussian variables lack that coefficient representation. **MGFs and CGFs are the main tools for Gaussian moments and linear combinations**; PGFs describe the random count N. In the next part, we derive the Gaussian MGF *from first principles* by completing the square rather than memorizing it.

5.7 Six Common Generating Functions at a Glance

Distribution and parameters	**MGF $M(t)$**	**PGF $G(s)$**	**Mean / variance**
Bernoulli(p)	$1-p+pe^t$	$1-p+ps$	p, $p(1-p)$
Binomial(n,p)	$(1-p+pe^t)^n$	$(1-p+ps)^n$	np, $np(1-p)$
Poisson(λ)	$e^{\lambda(e^t-1)}$	$e^{\lambda(s-1)}$	λ, λ
Exponential(λ)	$\lambda/(\lambda-t)$, $t<\lambda$	Not applicable	$1/\lambda$, $1/\lambda^2$
Normal(μ,σ^2)	$e^{\mu t+\sigma^2 t^2/2}$	Not applicable	μ, σ^2
Chi-square(ν)	$(1-2t)^{-\nu/2}$, $t<1/2$	Not applicable	ν, 2ν

How to use the table: The first distributions have already been derived. The normal and chi-square MGFs preview the next two chapters and serve as a *review index*; their full derivations appear there.

Why is Cauchy absent? For $X \sim$ Cauchy, its MGF generally diverges for nonzero t, and $\mathbb{E}[X]$ does not exist. **Differentiating a generating function requires existence conditions; it cannot be applied blindly to every distribution.**

6 The Key Bridge: Two Equivalent Methods for a Random Sum

6.1 Setup and Essential Independence Assumptions

Let $N \in \{0, 1, 2, \ldots\}$ be a count. Per-trade payoffs $R_1, R_2, \ldots$ are i.i.d., and **the entire payoff sequence is independent of** N. Define

$$S = \sum_{i=1}^{N} R_i, \qquad \mu = \mathbb{E}[R_1], \quad \sigma^2 = \text{Var}(R_1), \quad m = \mathbb{E}[N], \quad v = \text{Var}(N).$$

For $N = 0$, define the empty sum to be 0. We assume the required moments exist; the MGF proof additionally requires the relevant MGF to be finite in a neighborhood of zero.

6.2 Method A: Total Expectation and Total Variance Decompose Risk

After fixing $N = n$, the number of summands is no longer random. Therefore

$$\mathbb{E}[S \mid N = n] = n\mu, \qquad \text{Var}(S \mid N = n) = n\sigma^2.$$

Apply Tower first, and EVVE second:

$$\begin{aligned} \mathbb{E}[S] &= \mathbb{E}[\mathbb{E}[S \mid N]] = \mathbb{E}[N\mu] = m\mu, \\ \text{Var}(S) &= \mathbb{E}[\text{Var}(S \mid N)] + \text{Var}(\mathbb{E}[S \mid N]) \\ &= \mathbb{E}[N\sigma^2] + \text{Var}(N\mu) = \boxed{m\sigma^2 + v\mu^2}. \end{aligned}$$

> **Intuition and Memory**
>
> **Economic interpretation of EVVE:** $m\sigma^2$ measures *intrinsic per-trade risk*, which grows with the number of trades; $v\mu^2$ measures the *extra risk from random trade counts* whenever mean payoff $\mu \neq 0$. If $\mu = 0$, randomness in count still scales overall variance but does not create an extra "count $\times$ mean" variance term.

6.3 Method B: Compose the MGF with the PGF to Obtain the Same Answer

Transform one payoff into its MGF $M_R(t) = \mathbb{E}[e^{tR_1}]$, and the count into its PGF $G_N(s) = \mathbb{E}[s^N]$. Conditional on $N = n$, independence gives

$$\mathbb{E}[e^{tS} \mid N = n] = \prod_{i=1}^{n} \mathbb{E}[e^{tR_i}] = [M_R(t)]^n.$$

Now average over N (*this is still the Tower Property*):

$$\boxed{M_S(t) = \mathbb{E}\big[[M_R(t)]^N\big] = G_N(M_R(t)).}$$

First derivative: By the chain rule,

$$M_S'(t) = G_N'(M_R(t))\, M_R'(t).$$

Since $M_R(0) = 1$, $G_N'(1) = m$ and $M_R'(0) = \mu$,

$$\boxed{\mathbb{E}[S] = M_S'(0) = G_N'(1) M_R'(0) = m\mu.}$$

Second derivative: Remember the *two terms* in the chain rule:

$$\begin{aligned} M_S''(t) &= G_N''(M_R(t))[M_R'(t)]^2 \\ &\quad + G_N'(M_R(t))M_R''(t), \\ \mathbb{E}[S^2] &= G_N''(1)\mu^2 + G_N'(1)(\sigma^2+\mu^2). \end{aligned}$$

Substitute $G_N''(1) = \mathbb{E}[N(N-1)]$ and subtract the squared mean:

$$\begin{aligned} \text{Var}(S) &= m(\sigma^2+\mu^2) + \mathbb{E}[N(N-1)]\mu^2 - m^2\mu^2 \\ &= m\sigma^2 + \mu^2\big(\mathbb{E}[N^2] - m^2\big) \\ &= \boxed{m\sigma^2 + v\mu^2}. \end{aligned}$$

Why do both methods agree?
Conditioning separates the risk of S into *random payoffs* and a *random number of payoffs*; the generating-function method encodes exactly the same split through $G_N'(1)$, $G_N''(1)$, $M_R'(0)$ and $M_R''(0)$. The second PGF derivative is a falling-factorial moment; we must first "add the mean back" to reconstruct $\text{Var}(N)$. The proofs are the *probabilistic* and *algebraic* expressions of the same conditioning idea.

6.4 Compound Poisson: CGFs Give Variance and Higher Moments Faster

For $N \sim \text{Poisson}(\lambda)$, the previously derived PGF is

$$G_N(s) = e^{\lambda(s-1)} \quad \Longrightarrow \quad \boxed{M_S(t) = \exp\{\lambda[M_R(t)-1]\}.}$$

The random sum therefore has CGF $K_S(t) = \lambda[M_R(t)-1]$. Differentiation gives

$$\begin{aligned} \mathbb{E}[S] &= K_S'(0) = \lambda M_R'(0) = \lambda\mu, \\ \text{Var}(S) &= K_S''(0) = \lambda M_R''(0) = \boxed{\lambda(\sigma^2+\mu^2)}. \end{aligned}$$

More generally, whenever the raw moments exist, the rth cumulant of a compound Poisson sum is

$$\boxed{\kappa_r(S) = \lambda\mathbb{E}[R_1^r]}.$$

Important: This identity is special to compound Poisson sums. For a general random sum, it is not valid to assert $\kappa_r(S) = \mathbb{E}[N]\kappa_r(R)$; randomness in the number of terms introduces additional higher-order contributions.

6.5 Second Archetype: Verify the General Formula with Binomial Counts

Let $N \sim \text{Binomial}(n,p)$ be independent of all R_i, with $m = np$ and $v = np(1-p)$. Conditioning immediately gives

$$\boxed{\mathbb{E}[S] = np\mu, \qquad \text{Var}(S) = np\sigma^2 + np(1-p)\mu^2.}$$

The generating-function method needs no new formula:

$$\boxed{M_S(t) = \big[1-p+pM_R(t)\big]^n}.$$

Its first and second derivatives at $t=0$ yield the same result. If $R_i \sim \mathcal{N}(\mu,\sigma^2)$, substitute the Gaussian MGF derived later, $M_R(t) = e^{\mu t+\sigma^2t^2/2}$. But *do not* conclude that a random sum is normal merely because individual payoffs are normal.

6.6 Which Method Should You Choose?

Desired result	Preferred tool	Reason
Mean only	Tower / Wald-type sum	Fixing N immediately gives $\mu\mathbb{E}[N]$
Variance only	EVVE	Displays the two risk sources, $m\sigma^2 + v\mu^2$
Full transform / distribution	$M_S(t) = G_N(M_R(t))$	The random count becomes functional composition
Compound Poisson moments	$K_S(t) = \lambda(M_R(t) - 1)$	CGF derivatives directly give cumulants

If an MGF does not exist, the random sum can still be analyzed. The characteristic function $\varphi_R(u) = \mathbb{E}[e^{iuR}]$ always exists, and $\varphi_S(u) = G_N(\varphi_R(u))$. **MGF existence must be checked before relying on derivative arguments.**

7 Part III: Normal Distributions as a Central MGF / CGF Application

> **Intuition and Memory**
>
> **From tool to application:** The previous part established three rules: "derivatives generate moments", "independent sums become products", and "random counts become functional compositions". Rather than treat the Gaussian MGF as an isolated formula, we first *complete the square in the normal density* to derive M_Z, then obtain means, variances, even higher moments, independent normal sums, and lognormal price moments. PGFs do not apply directly to continuous normal variables, but the random-count model combines a PGF with a normal MGF.

7.1 Every Normal Variable Is an Affine Transform of a Standard Normal

Let $Z \sim \mathcal{N}(0,1)$. Then

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad \Phi(z) = \int_{-\infty}^{z} \phi(u)\,\mathrm{d}u, \qquad \Phi(-z) = 1 - \Phi(z).$$

For $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$, $X \overset{d}{=} \mu + \sigma Z$, so

$$\Pr(X > a) = 1 - \Phi\left(\frac{a-\mu}{\sigma}\right), \quad \Pr(a < X < b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

Common confusion: The second parameter of $\mathcal{N}(\mu, \sigma^2)$ is the *variance*, not the standard deviation; standardization divides by σ.

7.2 Derive the Normal MGF by Completing the Square

The MGF is $M_X(t) = \mathbb{E}[e^{tX}]$. For a standard normal, complete the square:

$$\begin{aligned} M_Z(t) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(tz - \frac{z^2}{2}\right) \mathrm{d}z \\ &= \exp\left(\frac{t^2}{2}\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(z-t)^2}{2}\right] \mathrm{d}z = e^{t^2/2}. \end{aligned}$$

The remaining integral is the full density of $\mathcal{N}(t,1)$, hence has area 1. Therefore

$$\boxed{M_X(t) = \mathbb{E}[e^{t(\mu+\sigma Z)}] = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right), \quad K_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2.}$$

Intuition and Memory

Two building blocks of the normal MGF: μt controls the location, and $\frac{1}{2}\sigma^2 t^2$ controls dispersion. Completing the square naturally produces $t^2/2$; there is no need to memorize its sign or coefficient separately.

7.3 Application 1: Derive the Normal Mean and Variance from MGF / CGF

We already proved $K_X'(0) = \mathbb{E}[X]$ and $K_X''(0) = \mathrm{Var}(X)$. Substituting the Gaussian CGF just derived requires *no further integration*:

$$\boxed{\mathbb{E}[X] = K_X'(0) = \mu, \qquad \mathrm{Var}(X) = K_X''(0) = \sigma^2.}$$

The two parameters are not additional results to memorize; they emerge directly from the MGF / CGF moment-generating mechanism.

7.4 Application 2: Sums of Independent Normal Variables

If the variables $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ are *independent*, then

$$\begin{aligned} M_{\sum_i X_i}(t) &= \prod_i M_{X_i}(t) \\ &= \exp\left(t\sum_i \mu_i + \frac{t^2}{2}\sum_i \sigma_i^2\right). \end{aligned}$$

Therefore

$$\boxed{\sum_{i=1}^n X_i \sim \mathcal{N}\left(\sum_i \mu_i, \sum_i \sigma_i^2\right).}$$

If the components are *jointly Gaussian but dependent*, linear combinations are still normal, but their variances must include covariance terms:

$$\mathrm{Var}\left(\sum_i a_i X_i\right) = \sum_i a_i^2 \sigma_i^2 + 2\sum_{i<j} a_i a_j \,\mathrm{Cov}(X_i, X_j).$$

For independent X, Y, $\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$. Squaring the minus sign still makes variance add.

7.5 Application 3: Lognormal Price Moments

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and set $S = e^X$, a lognormal variable. Evaluate the MGF at $t = 1$ and $t = 2$:

$$\begin{aligned} \mathbb{E}[S] &= M_X(1) = e^{\mu+\sigma^2/2}, \\ \mathbb{E}[S^2] &= M_X(2) = e^{2\mu+2\sigma^2}, \\ \boxed{\mathrm{Var}(S)} &= e^{2\mu+\sigma^2}(e^{\sigma^2} - 1). \end{aligned}$$

Application: Under normally distributed log returns (as in geometric Brownian motion at a specified time), the arithmetic price mean is not simply e^{μ}; it includes the positive $\sigma^2/2$ correction. The *mean log-price* is not the *logarithm of the mean price*.

7.6 Application 4: Higher Standard-Normal Moments and Double Factorials

Compare coefficients in $e^{t^2/2} = \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k!}$ and the moment expansion $M_Z(t) = \sum_{m=0}^{\infty} \frac{\mathbb{E}[Z^m]}{m!} t^m$:

$$\boxed{\mathbb{E}[Z^{2k+1}] = 0, \qquad \mathbb{E}[Z^{2k}] = (2k-1)!! = \frac{(2k)!}{2^k k!}.}$$

$$\mathbb{E}[Z^2] = 1, \quad \mathbb{E}[Z^4] = 3, \quad \mathbb{E}[Z^6] = 15, \quad \mathbb{E}[Z^8] = 105.$$

Intuition and Memory

Even moments are products of odd integers: Second moment 1, fourth moment 1×3, sixth moment $1 \times 3 \times 5$, and multiply by 7 for the eighth. Normal kurtosis is $\mathbb{E}[Z^4]/\operatorname{Var}(Z)^2 = 3$, with excess kurtosis 0.

7.7 Stein's Lemma: Replace Products by Derivative Expectations

The standard-normal density satisfies $\phi'(z) = -z\phi(z)$. For a differentiable function g, assuming integrability and vanishing integration-by-parts boundary terms,

$$\begin{aligned}\mathbb{E}[Zg(Z)] &= \int z g(z)\phi(z)\,\mathrm{d}z = -\int g(z)\phi'(z)\,\mathrm{d}z \\ &= \boxed{\mathbb{E}[g'(Z)]}.\end{aligned}$$

For example, with $g(z) = z^3$, obtain $\mathbb{E}[Z^4] = 3\mathbb{E}[Z^2] = 3$. For jointly normal (X, Y),

$$\boxed{\operatorname{Cov}(X, g(Y)) = \operatorname{Cov}(X, Y)\mathbb{E}[g'(Y)]}.$$

In particular, when standard jointly normal variables have correlation ρ,

$$\operatorname{Cov}(X, Y^3) = \rho\,\mathbb{E}[3Y^2] = 3\rho.$$

Intuition: In Gaussian models, the link between X and Y is compressed into the coefficient $\operatorname{Cov}(X, Y)$, while differentiation handles the more complicated $g(Y)$. This is integration by parts in probabilistic form.

7.8 Connection to Part II: A Random Number of Normal Payoffs Is Usually Not Normal

Let $N \sim \text{Poisson}(\lambda)$ with $\lambda > 0$, independent of all R_i. If $R_i \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$, then conditional on $N = n$,

$$S \mid N = n \sim \mathcal{N}(n\mu, n\sigma^2) \quad (n \geq 1), \qquad S \mid N = 0 = 0.$$

Unconditionally, we have a *mixture* of Gaussian conditional laws, and moreover,

$$\Pr(S = 0) \geq \Pr(N = 0) = e^{-\lambda} > 0.$$

Therefore, for $\lambda > 0$ and $\sigma > 0$, S cannot be a nondegenerate normal variable. The PGF–MGF composition from Part II gives its MGF:

$$\boxed{M_S(t) = \exp\left\{\lambda\left[e^{\mu t + \sigma^2 t^2/2} - 1\right]\right\}.}$$

> **Intuition and Memory**
>
> **Two easily confused addition rules:** A sum of a fixed number n of independent normals *is always normal*; a sum of a random number N of normals is *generally not normal*. Random counts create a mixture, exactly where Tower, EVVE, and generating functions become useful together.

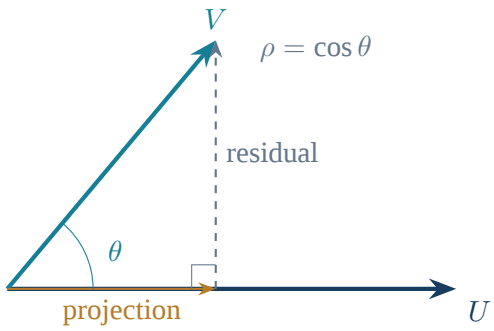
8 Gaussian Geometry: Correlation, Independence, and Conditional Prediction

8.1 Variance as Squared Length, Covariance as Inner Product, Correlation as Angle

For centered variables $U = X - \mathbb{E}[X]$ and $V = Y - \mathbb{E}[Y]$, define $\|U\|_2 = \sqrt{\mathbb{E}[U^2]}$. Then

$$\underbrace{\mathrm{Var}(X)}_{\text{squared length}} = \|U\|_2^2, \qquad \underbrace{\mathrm{Cov}(X,Y)}_{\text{inner product}} = \langle U, V\rangle, \qquad \underbrace{\rho}_{\text{cosine of angle}} = \frac{\langle U, V\rangle}{\|U\|_2 \|V\|_2}.$$

Cauchy–Schwarz gives $|\rho| \le 1$. At $\rho = 0$, the two centered random variables are orthogonal.



> **Common Pitfalls**
>
> For *arbitrary distributions*, $X \perp\!\!\!\perp Y \Rightarrow \mathrm{Cov}(X,Y) = 0$ when variances exist; the converse usually fails. **Jointly normal** variables satisfy $\mathrm{Cov}(X,Y) = 0 \iff X \perp\!\!\!\perp Y$. Geometric orthogonality does not automatically imply probabilistic independence; this is a special Gaussian property.

8.2 The Essential Counterexample: Normal Marginals Do Not Imply Joint Normality

Let $X \sim \mathcal{N}(0,1)$ and let an independent random sign B equal ± 1 with probability $1/2$ each. Define $Y = BX$. Then

$$Y \sim \mathcal{N}(0,1), \qquad \mathrm{Cov}(X,Y) = \mathbb{E}[BX^2] = \mathbb{E}[B]\mathbb{E}[X^2] = 0.$$

But $|Y| = |X|$ always, so X, Y are clearly dependent. Also,

$$\Pr(X + Y = 0) \ge \Pr(B = -1) = \frac{1}{2},$$

Thus $X + Y$ has a point mass and cannot be nondegenerate normal. **Conclusion: normal marginals plus zero covariance do not suffice for independence or for normality of the sum.**

8.3 How to Recognize Truly Jointly Gaussian Variables

For a random vector $\mathbf{X} = (X_1, \ldots, X_n)^\top$, the core definition of *joint Gaussianity* is that every linear combination $\mathbf{a}^\top \mathbf{X}$ is univariate normal (allowing a degenerate constant). Equivalently, it can be constructed as

$$\boxed{\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z}}, \qquad \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I), \quad \Sigma = AA^\top.$$

When the covariance matrix Σ is positive definite, an ordinary multivariate Gaussian density exists; when Σ is singular, the vector may still be jointly Gaussian but have no ordinary density on the whole ambient space. Thus **the density formula is not the fundamental test**.

8.4 The Universal Orthogonal Decomposition for Two Jointly Normal Variables

Start with the simplest standardized case: jointly normal (X, Y) with $X, Y \sim \mathcal{N}(0, 1)$ and $\text{Corr}(X, Y) = \rho$. Write

$$\boxed{Y = \rho X + \sqrt{1 - \rho^2}\, Z}, \qquad Z \sim \mathcal{N}(0, 1), \quad Z \perp\!\!\!\perp X \quad (|\rho| < 1).$$

This expresses the projection on X with coefficient ρ, plus remaining Gaussian noise independent of X. Check:

$$\mathbb{E}[Y] = 0, \quad \text{Var}(Y) = \rho^2 + (1 - \rho^2) = 1, \quad \text{Cov}(X, Y) = \rho.$$

More generally, if $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ are jointly normal, then

$$\boxed{Y = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X) + \sigma_Y \sqrt{1 - \rho^2}\, Z}, \qquad Z \perp\!\!\!\perp X.$$

> **Intuition and Memory**
>
> **One-sentence memory device:** Standardize into two unit-length vectors; *projection* removes the fraction ρ; *Pythagoras* gives an independent residual with scale $\sqrt{1 - \rho^2}$. Only after squaring do you obtain residual *variance* $1 - \rho^2$.

8.5 Bivariate Normal Conditioning: OLS Prediction Is Exact

Conditional on $X = x$, the residual Z above is unaffected and remains standard normal. Therefore

$$\boxed{\begin{aligned} Y \mid X = x &\sim \mathcal{N}\left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X),\ \sigma_Y^2(1 - \rho^2)\right), \\ \mathbb{E}[Y \mid X = x] &= \mu_Y + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(x - \mu_X), \\ \text{Var}(Y \mid X = x) &= \sigma_Y^2(1 - \rho^2). \end{aligned}}$$

Swapping X and Y gives the formula for $X \mid Y = y$. **Conditional variance does not depend on the observed value x:** A larger positive signal shifts the predicted center but does not change residual uncertainty under this jointly normal model.

Standard interview question: For a standard bivariate normal pair with correlation ρ,

$$X \mid Y = y \sim \mathcal{N}(\rho y, 1 - \rho^2), \qquad \boxed{\Pr(X > 0 \mid Y = y) = \Phi\left(\frac{\rho y}{\sqrt{1-\rho^2}}\right)}$$

(The conditional-probability expression assumes $|\rho| < 1$; for $\rho = \pm 1$, apply the deterministic relation.)

8.6 Sum and Difference: Rotational Independence for Equal-Variance Joint Normals

Set $U = X + Y$ and $V = X - Y$. Regardless of whether X, Y are correlated,

$$\begin{aligned}\text{Cov}(U, V) &= \text{Cov}(X + Y, X - Y) \\ &= \text{Var}(X) - \text{Var}(Y).\end{aligned}$$

If $\text{Var}(X) = \text{Var}(Y)$ and (X, Y) are jointly normal, U, V are jointly normal and uncorrelated; hence

$$\boxed{X + Y \perp\!\!\!\perp X - Y \qquad \text{(joint Gaussian and equal variances)}.}$$

Geometric memory: For two equal-length vectors, the *sum* and *difference* are the diagonals of a rhombus and are perpendicular. Joint normality upgrades perpendicularity to independence.

9 Multivariate Normal: Matrices, Correlation Bounds, and Signal Extraction

9.1 The Matrix Form Packages All One-Dimensional Rules

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, where Σ is symmetric positive semidefinite. Then

$$\boxed{\mathbf{Y} = A\mathbf{X} + \mathbf{b} \sim \mathcal{N}(A\boldsymbol{\mu} + \mathbf{b}, A\Sigma A^\top)}.$$

For any vector $\mathbf{a}$, the mean and variance of $\mathbf{a}^\top \mathbf{X}$ are $\mathbf{a}^\top \boldsymbol{\mu}$ and $\mathbf{a}^\top \Sigma \mathbf{a}$, respectively. This is the probability basis for linear portfolio return models and portfolio risk $w^\top \Sigma w$. $\mathbf{a}^\top \boldsymbol{\mu}$ and $\mathbf{a}^\top \Sigma \mathbf{a}$, respectively. These are the direct probabilistic basis of linear portfolio-return models and portfolio risk $w^\top \Sigma w$.

9.2 Multivariate Gaussian Conditioning = Multivariate OLS / Optimal Linear Filtering

Partition a jointly Gaussian vector as $\binom{\mathbf{X}}{\mathbf{Y}}$, with mean $\binom{\mu_X}{\mu_Y}$ and covariance

$$\Sigma = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix}.$$

If Σ_{YY} is invertible, then

$$\boxed{\begin{aligned}\mathbb{E}[\mathbf{X} \mid \mathbf{Y} = \mathbf{y}] &= \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(\mathbf{y} - \mu_Y), \\ \text{Cov}(\mathbf{X} \mid \mathbf{Y}) &= \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}.\end{aligned}}$$

Remember: Prediction = prior mean + "covariance regression coefficient" $\times$ new information. Remaining risk = original risk $-$ risk explained by the observations (Schur complement). The one-dimensional special case contains $\rho\sigma_X/\sigma_Y$ and $\sigma_X^2(1-\rho^2)$.

9.3 Noisy Alpha: Signal-to-Noise Shrinkage

This is the zero-mean special case of continuous Bayes updating in Section 2.4. We can use covariance projection directly or multiply the Prior and Likelihood and complete the square to obtain the same shrinkage weight.

Problem: Latent return $X \sim \mathcal{N}(0, \sigma_X^2)$, independent observation noise $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, signal $Y = X + \varepsilon$, with observed $Y = y$.

$$\text{Var}(Y) = \sigma_X^2 + \sigma_\varepsilon^2, \qquad \text{Cov}(X, Y) = \sigma_X^2.$$

Therefore

$$\boxed{\begin{aligned} \mathbb{E}[X \mid Y = y] &= \underbrace{\frac{\sigma_X^2}{\sigma_X^2 + \sigma_\varepsilon^2}}_{\text{reliability / shrinkage weight}} y, \\ \text{Var}(X \mid Y = y) &= \frac{\sigma_X^2 \sigma_\varepsilon^2}{\sigma_X^2 + \sigma_\varepsilon^2}. \end{aligned}}$$

As noise vanishes, the weight tends to 1 and we trust the observation; as noise diverges, the weight tends to 0 and the prediction returns to prior mean 0. **This is the static prototype of Gaussian Bayesian updating / the Kalman gain.**

9.4 Three-Asset Correlation Bounds from Positive Semidefiniteness

Problem: Given $\text{Corr}(X, Y) = r_1$ and $\text{Corr}(Y, Z) = r_2$, let $\text{Corr}(X, Z) = r_3$. Find its feasible range. The correlation matrix

$$R = \begin{pmatrix} 1 & r_1 & r_3 \\ r_1 & 1 & r_2 \\ r_3 & r_2 & 1 \end{pmatrix}$$

must be positive semidefinite, so its determinant is nonnegative:

$$1 - r_1^2 - r_2^2 - r_3^2 + 2r_1 r_2 r_3 \geq 0.$$

Solve this quadratic inequality for r_3:

$$\boxed{r_1 r_2 - \sqrt{(1 - r_1^2)(1 - r_2^2)} \leq r_3 \leq r_1 r_2 + \sqrt{(1 - r_1^2)(1 - r_2^2)}.}$$

Geometric view: Write $r_j = \cos\theta_j$; the extrema come from sums or differences of vector angles and the cosine rule. **Matrix view:** A valid covariance structure makes every portfolio variance nonnegative. Both views express the same condition.

9.5 Equicorrelated Portfolios Cannot Have Arbitrarily Negative Correlations

If n standardized assets have common pairwise correlation r, then

$$\text{Var}\Big(\sum_{i=1}^{n} X_i\Big) = n + n(n-1)r \geq 0.$$

Together with $r \leq 1$, this gives the exact feasible interval for an equicorrelation matrix:

$$\boxed{-\frac{1}{n-1} \leq r \leq 1.}$$

For example, three assets cannot *all* have correlation -0.8; the minimum is -0.5. The eigenvalues of their correlation matrix are $1-r$ with multiplicity $n-1$, and $1+(n-1)r$; both must be nonnegative.

10 Gaussian Interview Applications: Sums, Squares, Angles, Ratios, and Higher Moments

10.1 Linear Combinations Stay Gaussian; Squares Generally Do Not

The closure property of jointly normal distributions applies to **linear transformations**:

$$aX+bY \sim \mathcal{N}\big(a\mu_X+b\mu_Y,\ a^2\sigma_X^2+b^2\sigma_Y^2+2ab\rho\sigma_X\sigma_Y\big).$$

But X^2, X/Y and $\max(X,Y)$ are nonlinear transformations and generally *not normally distributed*. Common interview patterns are **squaring → chi-square; ratios → Cauchy; maximum → an absolute-value identity**.

10.2 Sums of Normal Squares: χ^2 and Its MGF

We *reuse the MGF product rule from Part II*. Note carefully: $M_{Z^2}(t)=\mathbb{E}[e^{tZ^2}]$ is not $M_Z(t)$ squared. **The MGF of the squared variable must be derived anew**, before applying independence to a sum of n squares. For $Z\sim\mathcal{N}(0,1)$, $Z^2\sim\chi_1^2$. Normal moments give

$$\mathbb{E}[Z^2]=1, \qquad \mathrm{Var}(Z^2)=\mathbb{E}[Z^4]-(\mathbb{E}[Z^2])^2=3-1=2.$$

If $Z_1,\dots,Z_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$, then

$$\boxed{Q=\sum_{i=1}^{n} Z_i^2\sim\chi_n^2, \qquad \mathbb{E}[Q]=n, \quad \mathrm{Var}(Q)=2n.}$$

To derive the MGF, integrate directly with respect to the density of Z:

$$\begin{aligned} M_{Z^2}(t)=\mathbb{E}[e^{tZ^2}] &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(1-2t)z^2/2}\,\mathrm{d}z \\ &= (1-2t)^{-1/2}, \quad t<\frac{1}{2}. \end{aligned}$$

The independent product gives

$$\boxed{M_{\chi_n^2}(t)=(1-2t)^{-n/2}, \qquad t<\frac{1}{2}.}$$

With Gamma written in *shape/scale* notation $\Gamma(k,\theta)$, $\chi_n^2=\Gamma(n/2,2)$; under the *shape/rate* convention it is $\Gamma(n/2,1/2)$. **Always check what the Gamma distribution's second parameter means.**

10.3 Why Are the Sample Mean and Sample Variance Independent?

Let $X_1,\dots,X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu,\sigma^2)$, and define

$$\overline{X}=\frac{1}{n}\sum_i X_i, \qquad S^2=\frac{1}{n-1}\sum_i (X_i-\overline{X})^2.$$

The sample mean is the one-dimensional projection *along the all-ones vector*; the residuals $(X_i - \overline{X})_{i=1}^n$ lie in the orthogonal $(n-1)$-dimensional subspace.

Step 1: Calculate the covariance.

$$\mathrm{Cov}(\overline{X}, X_i - \overline{X}) = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0.$$

Because the mean and the entire residual vector are *jointly normal*, zero cross-covariance makes $\overline{X}$ independent of that vector, and hence independent of its function S^2.

Step 2: Count dimensions. The sum of squares of n standard-normal coordinates decomposes into one squared coordinate along the mean direction and $n-1$ squared coordinates in the orthogonal residual space. Thus,

$$\boxed{\overline{X} \perp\!\!\!\perp S^2, \qquad \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n} \left(\frac{X_i - \overline{X}}{\sigma}\right)^2 \sim \chi^2_{n-1}.}$$

Intuition and Memory

Where do the $n-1$ degrees of freedom come from? The residuals must sum to zero: $\sum_i (X_i - \overline{X}) = 0$, removing one independent direction. **The sample mean takes one dimension; the sample variance uses the other $n-1$.** This independence relies on joint Gaussianity and cannot automatically be extended to arbitrary i.i.d. distributions.

10.4 The Maximum of Two Normals: Rewrite It Using Absolute Value

Problem: Let (X, Y) be jointly normal with $\mathbb{E}[X] = \mathbb{E}[Y] = \mu$, $\mathrm{Var}(X) = \mathrm{Var}(Y) = \sigma^2$, correlation ρ. Find $\mathbb{E}[\max(X, Y)]$. The following identity holds for any two real numbers:

$$\boxed{\max(a, b) = \frac{a + b + |a - b|}{2}}.$$

First calculate the folded-normal expectation:

$$\mathbb{E}[|Z|] = 2\int_0^\infty z\phi(z)\,\mathrm{d}z = \sqrt{\frac{2}{\pi}}.$$

Next, set $D = X - Y \sim \mathcal{N}(0, 2\sigma^2(1-\rho))$. Then

$$\begin{aligned}\mathbb{E}[\max(X, Y)] &= \mu + \frac{1}{2}\mathbb{E}[|D|] \\ &= \mu + \frac{1}{2}\sqrt{2\sigma^2(1-\rho)}\sqrt{\frac{2}{\pi}} \\ &= \boxed{\mu + \sigma\sqrt{\frac{1-\rho}{\pi}}}.\end{aligned}$$

Limit check: At $\rho = 1$, the two variables agree almost surely and the extra maximum premium is zero. At $\rho = -1$, their difference is most volatile and the maximum premium is largest.

10.5 Same-Sign Probabilities for a Standard Bivariate Normal: Use Angles

Problem: Jointly standard-normal (X, Y) with $\mathrm{Corr}(X, Y) = \rho$. Find $\Pr(X > 0, Y > 0)$ and $\Pr(XY > 0)$.

$$Y = \rho X + \sqrt{1-\rho^2}\,Z, \qquad X \perp\!\!\!\perp Z \sim \mathcal{N}(0, 1).$$

The density of the two-dimensional vector (X, Z) depends only on $x^2 + z^2$, so its polar angle Θ is *uniform* around the full circle. Let $\alpha = \arcsin \rho$. In polar coordinates,

$$X = R\cos\Theta, \qquad Y = R\sin(\Theta + \alpha).$$

The angular overlap of the regions $X > 0$ and $Y > 0$ is $\pi/2 + \alpha$. Therefore,

$$\boxed{\Pr(X > 0, Y > 0) = \frac{1}{4} + \frac{\arcsin\rho}{2\pi}}, \qquad \boxed{\Pr(XY > 0) = \frac{1}{2} + \frac{\arcsin\rho}{\pi}}.$$

Limit check: $\rho = 0 \Rightarrow \Pr(X > 0, Y > 0) = 1/4$; $\rho = 1 \Rightarrow 1/2$; $\rho = -1 \Rightarrow 0$. **The shortest interview explanation is not a "double integral", but "rotational symmetry of a two-dimensional normal plus the fraction of the circle".**

10.6 Correlation of Squared Returns: $\mathrm{Corr}(X^2, Y^2) = \rho^2$

Problem: For standard jointly normal variables with correlation ρ, find $\mathbb{E}[X^2Y^2]$, $\mathrm{Cov}(X^2, Y^2)$ and their squared-variable correlation.

Use the orthogonal decomposition $Y = \rho X + \sqrt{1-\rho^2}Z$ with X, Z independent standard normals:

$$\begin{aligned}\mathbb{E}[X^2Y^2] &= \rho^2\mathbb{E}[X^4] + 2\rho\sqrt{1-\rho^2}\mathbb{E}[X^3]\mathbb{E}[Z] + (1-\rho^2)\mathbb{E}[X^2]\mathbb{E}[Z^2] \\ &= 3\rho^2 + 0 + 1 - \rho^2 = 1 + 2\rho^2.\end{aligned}$$

Using $\mathbb{E}[X^2] = \mathbb{E}[Y^2] = 1$ and $\mathrm{Var}(X^2) = \mathrm{Var}(Y^2) = 2$,

$$\boxed{\mathrm{Cov}(X^2, Y^2) = 2\rho^2, \qquad \mathrm{Corr}(X^2, Y^2) = \rho^2.}$$

Intuition and Memory

Return correlation can be positive or negative; squared-return correlation in this Gaussian model is always nonnegative. For $\rho = -0.8$, it is 0.64. These are *single-period squared returns*: they illustrate the tendency of large movements to coincide, but should not be equated directly with empirical realized volatility or volatility clustering.

10.7 Ratio of Independent Standard Normals: Cauchy and Undefined Moments

Let $X, Y \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. Since a two-dimensional normal has uniform polar angle, $X/Y = \cot\Theta$ has the distribution of the tangent of a uniform angle. A change of variables gives

$$\boxed{W = \frac{X}{Y} \sim \text{Cauchy}(0,1), \qquad f_W(w) = \frac{1}{\pi(1+w^2)}, \quad w \in \mathbb{R}.}$$

Both $\mathbb{E}[W]$ and $\mathrm{Var}(W)$ *do not exist* (the first absolute moment diverges). Symmetry of f_W does not imply a mean of zero: the expectation itself is not integrable.

Polar-angle mini-problem: Find $\Pr(X > \sqrt{3}Y > 0)$. In the first quadrant $0 < \Theta < \pi/2$, the condition $X > \sqrt{3}Y$ is equivalent to $\tan\Theta < 1/\sqrt{3}$, or $0 < \Theta < \pi/6$. Thus,

$$\boxed{\Pr(X > \sqrt{3}Y > 0) = \frac{\pi/6}{2\pi} = \frac{1}{12}.}$$

Common Pitfalls

Do not turn "both variables are normal" into "every function is normal". Sums need independence or at least *joint normality*; squares produce χ^2; ratios can be Cauchy, and symmetry about zero does *not* justify writing $\mathbb{E}[X/Y] = 0$.

11 Quant Interviews: Identify the Problem Type and Choose the Shortest Method

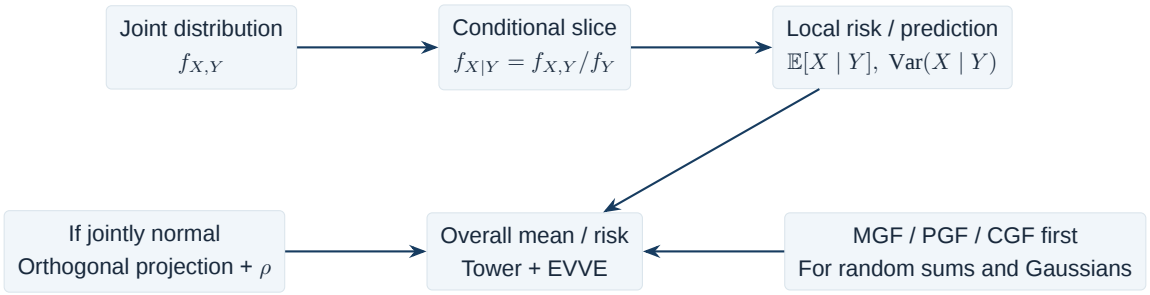
11.1 Keywords and the Shortest Mathematical Route

1. **"Given $Y = y$":** Identify target X and condition Y. If a joint density is supplied, divide by $f_Y(y)$; if a conditional rule is supplied, use it directly.
2. **"Overall probability / mean":** First condition to reduce dimension, multiply by the conditioning variable's probability mass, then integrate or sum.
3. **"Variance under random conditions":** Write EVVE immediately and separate variation *within conditions* from variation *between conditions*.
4. **"Gaussian":** Distinguish marginal from joint normality. Under joint Gaussianity, prefer projection / orthogonal decomposition to a hand calculation of the joint density.
5. **"Normal sums / squares":** Linear combinations remain Gaussian; independent standard-normal squares sum to χ^2; sample means and residuals split into orthogonal dimensions.
6. **"$\mathbb{E}[e^{tX}]$ / derivatives":** Decide whether the transform is an MGF, PGF, or CGF, then evaluate at 0 or 1 as appropriate.
7. **"Random number of trades":** First fix the count N, then apply Tower + EVVE. For a full transform, use $G_N(M_R(t))$.

11.2 Consistency Checks Every Answer Must Pass

- A conditional density must be nonnegative and integrate to one over its target; a marginal density must integrate to one on its proper support.
- All variances must be nonnegative, $|\operatorname{Corr}| \leq 1$, and correlation matrices positive semidefinite.
- Check limits such as $\rho = 0, \pm 1$, zero/infinite observation noise, and count intensity $\lambda \to 0$.
- For G and M, $G(1) = 1$ and $M(0) = 1$; for K, $K(0) = 0$.
- A continuous density may exceed one in height, but its integral must be one; a single point can still have zero probability.
- Every use of independence needs either an *explicit independence assumption* or a deduction from *joint normality with zero covariance*.

A Interview Quick Review I: A Unified Map of the Formulas



Fix the condition; divide by the marginal → take conditional x-moments → average over Y, weighted by p_Y or $f_Y\,\mathrm{d}y$.

A.1 The Six "Arrows" as Formula Templates

$$
\begin{aligned}
\text{Marginalization:} &\quad f_Y(y) = \int f_{X,Y}(x,y)\,\mathrm{d}x.\\
\text{Conditioning:} &\quad f_{X|Y}(x \mid y) = f_{X,Y}(x,y)/f_Y(y).\\
\text{Continuous Bayes:} &\quad f_{X|Y}(x \mid y) = \frac{f_X(x) f_{Y|X}(y \mid x)}{\int f_X(u) f_{Y|X}(y \mid u)\,\mathrm{d}u}.\\
\text{Marginal reconstruction:} &\quad f_X(x) = \int f_{X|Y}(x \mid y) f_Y(y)\,\mathrm{d}y.\\
\text{Mean reconstruction:} &\quad \mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].\\
\text{Variance reconstruction:} &\quad \mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]).\\
\text{Random-sum transform:} &\quad M_{\sum_{i=1}^{N} R_i}(t) = G_N(M_R(t)).
\end{aligned}
$$

A.2 A Static Memory Picture of a Conditional Normal Slice

Consider a standard bivariate normal with $\rho = 0.6$ and $Y = 2$:

$$
f_Y(2) = \phi(2) = \frac{e^{-2}}{\sqrt{2\pi}} \approx 0.0540, \qquad X \mid Y = 2 \sim \mathcal{N}(1.2, 0.64).
$$

The joint-density slice $f_{X,Y}(x,2) = f_Y(2) f_{X|Y}(x \mid 2)$ has area approximately 0.054; after dividing by $f_Y(2)$, the conditional density has area one. **The fact that $Y = 2$ is a relatively rare input does not mean the conditional density of X must be small after that input is observed.**

B Interview Quick Review II: Gaussian Formula Sheet

Topic	Formula and assumptions
Standardization	$Z = (X-\mu)/\sigma \sim \mathcal{N}(0,1)$, $\Phi(-z) = 1 - \Phi(z)$
MGF / CGF	$M_X(t) = e^{\mu t + \sigma^2 t^2/2}$, $K_X(t) = \mu t + \sigma^2 t^2/2$
Even moments	$\mathbb{E}[Z^{2k}] = (2k-1)!!$; $\mathbb{E}[Z^4] = 3,\ \mathbb{E}[Z^6] = 15,\ \mathbb{E}[Z^8] = 105$

Topic	Formula and conditions
Stein	$\mathbb{E}[Zg(Z)] = \mathbb{E}[g'(Z)]$; $\text{Cov}(X, g(Y)) = \text{Cov}(X, Y)\mathbb{E}[g'(Y)]$ (joint normal)
Linear combination	$aX + bY \sim \mathcal{N}(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y)$ (joint normal)
Correlation / independence	$X \perp\!\!\!\perp Y \iff \text{Cov}(X, Y) = 0$ (joint normal; marginal normality *alone is insufficient*)
Orthogonal decomposition	$Y = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(X - \mu_X) + \sigma_Y\sqrt{1-\rho^2}Z$, $Z \perp\!\!\!\perp X$
Conditional Gaussian	$Y \mid X = x \sim \mathcal{N}\big(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1-\rho^2)\big)$
Multivariate affine transform	$A\mathbf{X} + \mathbf{b} \sim \mathcal{N}(A\mu + \mathbf{b}, A\Sigma A^\top)$
Multivariate conditional mean	$\mu_{X\|y} = \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y)$
Multivariate conditional covariance	$\Sigma_{X\|Y} = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}$
Sum–difference independence	$X + Y \perp\!\!\!\perp X - Y$ (joint normal with equal variances)
Chi-square	$\sum_{i=1}^n Z_i^2 \sim \chi_n^2$ (independent standard normals); $\mathbb{E} = n$, $\text{Var} = 2n$
Chi-square MGF	$M_{\chi_n^2}(t) = (1-2t)^{-n/2}$, $t < 1/2$
Sample mean and variance	$\overline{X} \perp\!\!\!\perp S^2$; $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$ (normal i.i.d.)
Absolute value and maximum	$\mathbb{E}\|Z\| = \sqrt{2/\pi}$; $\mathbb{E}\max(X, Y) = \mu + \sigma\sqrt{(1-\rho)/\pi}$ (equal means and variances)
Gaussian same sign	$\Pr(X > 0, Y > 0) = 1/4 + \arcsin(\rho)/(2\pi)$ (standard joint normal)
Squared correlation	$\mathbb{E}[X^2Y^2] = 1 + 2\rho^2$, $\text{Corr}(X^2, Y^2) = \rho^2$ (standard joint normal)
Three-variable correlation bounds	$r_1r_2 \pm \sqrt{(1-r_1^2)(1-r_2^2)}$ bound the unknown correlation
Equicorrelation matrix	$-1/(n-1) \le r \le 1$
Noisy observation	$\mathbb{E}[X \mid X + \varepsilon = y] = \frac{\sigma_X^2}{\sigma_X^2+\sigma_\varepsilon^2}y$ (independent zero-mean Gaussians)
Ratio of two normals	$X/Y \sim \text{Cauchy}(0, 1)$ (independent standard normals); mean undefined

C Interview Quick Review III: Generating Functions and Problem Index

Derivation sequence: PGF / MGF / CGF ⟶ raw moments and cumulants ⟶ Gaussian high moments and independent sums. For random-count questions, check transform-derived results with Tower / EVVE in reverse.

$$G(s) = \mathbb{E}[s^X],\ s = 1: \quad G'(1) = \mathbb{E}[X], \quad G''(1) = \mathbb{E}[X(X-1)],$$
$$M(t) = \mathbb{E}[e^{tX}],\ t = 0: \quad M'(0) = \mathbb{E}[X], \quad M''(0) = \mathbb{E}[X^2],$$
$$K(t) = \ln M(t),\ t = 0: \quad K'(0) = \mathbb{E}[X], \quad K''(0) = \text{Var}(X).$$

First derivatives all give means; second derivatives differ: PGF = factorial moment, MGF = raw moment, CGF = variance. Evaluate G at 1 and M/K at 0. For variance, "PGF: add the mean and subtract its square; MGF: subtract the square; CGF: no adjustment".

Archetype	Recognition	Final answer
Joint exponential kernel	$2e^{-x-y}$, $0 < y < x$; shifted conditional Exp slice	$\mathbb{E}[X \mid Y = y] = y + 1$
Exponential race	Independent $X \sim \text{Exp}(\lambda_1), Y \sim \text{Exp}(\lambda_2)$	$\Pr(X < Y) = \lambda_1/(\lambda_1 + \lambda_2)$
Hierarchical uniform	$Y \sim U(0,1), X \mid Y \sim U(0,Y)$	$f_X(x) = -\ln x$; $\mathbb{E}X = 1/4$; $\text{Var}\, X = 7/144$
Random coin bias	$P \sim U(0,1)$; toss n times given P	All heads: $1/(n+1)$
Standard bivariate Gaussian	Joint Gaussian with correlation ρ	$X \mid Y = y \sim \mathcal{N}(\rho y, 1-\rho^2)$
Gaussian Bayes	$X \sim \mathcal{N}(\mu_0, \tau^2)$; $Y = X + \varepsilon$, noise variance s^2	$K = \tau^2/(\tau^2+s^2)$; $\mu_{\text{post}} = \mu_0 + K(y - \mu_0)$
Compound Poisson	$N \sim \text{Poisson}(\lambda)$, $S = \sum_{i=1}^{N} R_i$	$\mathbb{E}S = \lambda\mu$; $\text{Var}\, S = \lambda(\sigma^2 + \mu^2)$
Maximum of two normals	Equal mean μ, standard deviation σ, correlation ρ	$\mu + \sigma\sqrt{(1-\rho)/\pi}$
Gaussian same-sign probability	Standard joint normal; use uniform polar angle	$1/4 + \arcsin(\rho)/(2\pi)$
Squared-return correlation	Standard joint normal; independent-residual expansion	$\text{Corr}(X^2, Y^2) = \rho^2$
Three-asset correlation	Correlation matrix must be PSD	$r_1 r_2 \pm \sqrt{(1-r_1^2)(1-r_2^2)}$
Alpha denoising	$Y = X + \varepsilon$, independent Gaussians	Shrinkage $\sigma_X^2/(\sigma_X^2 + \sigma_\varepsilon^2)$
Gaussian ratios and sectors	Independent standard normals; uniform polar angle	Cauchy; $\Pr(X > \sqrt{3}Y > 0) = 1/12$

Five Final Memory Threads

1. **Divide or multiply:** Joint → conditional means divide by the marginal; conditional → global means multiply by conditional weights and aggregate. For a reverse observation mechanism, normalize Prior × Likelihood using Bayes.
2. **Total expectation and variance:** A mean averages across groups; variance is the average within-group risk plus variance of the between-group means (EVVE).

3. **Generating functions are tools:** Ask whether to evaluate at 0 or 1. PGF gives factorial moments, MGF gives raw moments, and CGF gives variance.
4. **Gaussian is an application:** Complete the square to derive the normal MGF, moments and independent sums; covariance is an inner product and conditional expectation is a projection.
5. **Two proofs of random sums:** Conditioning uses Tower + EVVE; transforms use two derivatives of $M_S(t) = G_N(M_R(t))$. Both give $\text{Var}(S) = \mathbb{E}[N]\sigma^2 + \text{Var}(N)\mu^2$.

ABOUT THE AUTHOR

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I'm Mizuki Seika, a quantitative trader working at the intersection of mathematics, financial markets, and emerging technology; an explorer of Web3 and cryptocurrency; and, in a parallel universe that happens to be equally real, an independent idol and original music producer.

My academic path spans Carnegie Mellon University's Master of Science in Computational Finance (MSCF '28), Mathematics at Tsinghua University ('25), and an exchange year at the University of Oxford ('24). My interests range from quantitative research and systematic trading to decentralized finance and digital assets.

To me, markets and music have more in common than they appear to: both reward an ear for signals, a sense of structure, and the courage to build something original.

> *In finance, I turn complexity into models.*
> *In music, I turn it into melodies.*

This work is where I bring that same creative discipline to learning --- reconstructing knowledge until the pieces form a framework that actually makes sense.

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Thank you for reading.